

Walking Through Time: A Hybrid Immersive System for Spatial and Expressive Timeline Authoring

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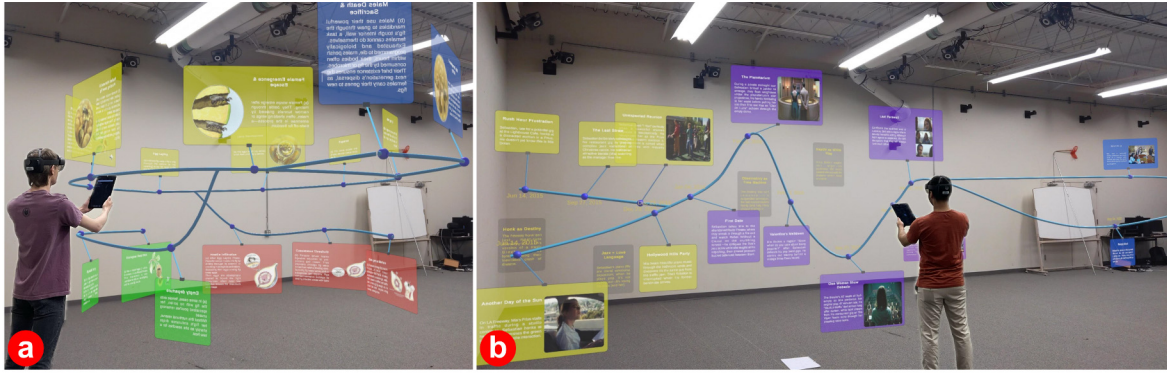


Figure 1: We present a timeline authoring tool that integrates augmented reality and tablet-based spatial interaction and explore its creative potential using two datasets: (a) the repetitive Fig wasp lifecycle, and (b) the La La Land's emotional narrative arc.

ABSTRACT

This paper introduces a timeline authoring tool that integrates Augmented Reality and tablet-based spatial interaction, enabling users to create large-scale non-linear timelines by leveraging spatial cognition and embodied interaction. In this system, the AR headset provides a large, immersive space for visualizing and interacting with a room-sized 3D timeline, while the tablet allows for precise sketching, annotation, and sculpting of timeline structures such as curves and branches. We conducted a design workshop to explore the user experience with this tool, which supports the creation of diverse, expressive timelines for two datasets.

Index Terms: Timeline, authoring, mobile, immersive analytics.

1 INTRODUCTION

Timeline visualization is one of the most fundamental and widely used visualization types for helping users understand temporal events [2]. However, traditional timeline authoring approaches remain constrained by two major limitations: (1) they rely on flat, linear representations (like single lines or Gantt charts) that fail to reflect how people often recall and understand events in non-linear forms, and (2) they follow strict authoring structures that prioritize exact time sequences over flexible, story-driven representations.

One possible solution to these problems is augmented reality (AR), which enables more immersive and flexible spatial representations of time-based events in expansive 3D space. AR enables intuitive spatial organization and understanding of temporal relationships, surpassing screen-based timelines.

However, it remains a significant challenge and a relatively underexplored area for users to create, edit, and explore large-scale timeline visualizations in immersive space.

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We present a new hybrid AR system that treats timeline authoring as a physical and spatial activity, bridging the gap between abstract representations of time and the more embodied way people experience it. It allows users to walk through time and revisit moments using embodied movement, transforming timelines from static 2D sequences into physically navigable 3D structures that align with natural memory processes (Figure 1).

Our work makes two primary contributions:

The hybrid timeline authoring tool combines tablet-based spatial interaction with immersive AR visualization, enabling users to sculpt large-scale 3D timelines through direct manipulation. It supports free-form shapes (e.g., curves, branches), embodied interaction (e.g., walking, head movement), and dynamic event editing/management, translating key narrativity principles [6] into interactive techniques.

Insights from an exploratory workshop demonstrate how the spatial interaction influences both timeline authoring and interpretation of time. The results revealed common strategies and processes for creating large-scale timelines. Our presented approach enhanced users' understanding of event relationships through physical navigation, while 2D-like timelines provided clearer overviews.

2 RELATED WORK

We situate our work at the intersection of timeline authoring and embodied hybrid tablet-AR interaction. This approach transforms traditional linear narratives into flexible 3D structures that support both creative expression and spatial memory.

2.1 Timeline Authoring

Existing research has explored various authoring approaches to timeline creation, visualization, and interaction, focusing on enhancing user understanding and engagement with temporal data. Prior studies introduced tools that support linear [2, 4], non-linear formats such as circular and spiral timelines [27, 2, 4, 5], and free-form sketching techniques [21, 5]. Free-form sketching has been shown to support sensemaking through metaphorical or figurative mnemonic cues [26, 17]. We extend this concept by enabling users to sculpt large-scale timelines in 3D space using a tablet, while introducing emotional mapping—where steep curves represent tur-

bulence and gentle slopes convey calm—to spatially encode subjective experiences.

Some systems have automated timeline generation from structured data such as spreadsheets [9, 14, 23], but these often limited creative flexibility. In contrast, our system emphasizes manual, spatially grounded authoring, allowing users to iteratively construct timelines using the tablet’s spatial interaction. This approach balances structure and creativity, making it useful for personal or hypothetical timelines where individual perspective and interpretation of timelines play a central role [25].

Although many timeline systems emphasized narrative structure, they often lacked an immersive and spatial context that allows users to physically navigate and experience the timeline. Prior work highlighted the potential of immersive environments for improving data comprehension [19, 12, 15] and supporting engaging storytelling [31, 20]. To leverage the potential explored in prior work, we combine free-form sketching with hybrid tablet-AR interaction, anchoring events to physical locations in 3D space.

2.2 Hybrid Interaction for Temporal Representation

Building on established connections between physical interaction and cognitive engagement, prior research has demonstrated how kinesthetic navigation reinforces spatial memory in museum contexts [28] and improves task performance compared to virtual navigation [3]. These findings align with Isenberg’s conceptual framework for immersive storytelling, which emphasized deep cognitive absorption through embodied experience [16]. However, implementations such as the glyph-based narratives [22], 3DStoryline visual storytelling [30], and environmental navigation [11] focused primarily on consumption rather than content creation.

To bridge physical and digital storytelling, we propose a tablet-AR hybrid system that uses device movement as both an input method and an authoring tool, enabling users to visualize narratives in real-world space [32, 13]. This builds on prior work in tablet-AR sensemaking, where systems like CHES [28] aligned phone-based AR with museum exhibits to embed narrative hotspots, and MAR-VisT [34] allowed users to reposition media by physically dragging their phones in 3D space. Additional studies [18, 33, 24] explored tablet-based AR/VR interactions for object manipulation and glyph visualization, but these systems primarily focused on spatial organization rather than temporal structuring for the creation of timelines.

Our work extends prior systems by integrating the precision input and physicality of tablets with AR’s spatial affordances while introducing direct timeline geometry manipulation. The tablet-AR hybrid can address limitations of mid-air interactions, such as imprecise hand tracking and drag-and-drop errors [29, 8], by leveraging the tablet’s interaction affordances and touch input. Specifically, the tablet provides a stable physical reference, allowing users to intuitively place and update event or annotation cards, rather than manipulating them with repeated finger gestures. This enables users to adopt more comfortable arm postures at natural lap or desk height during extended use. By unifying spatial and temporal authoring of timelines across AR and mobile devices, our approach offers a more holistic tool for temporal storytelling.

3 DESIGN CONSIDERATIONS

Our authoring tool is inspired by the work of exhibition designers who create large physical timelines to help visitors understand how time unfolds for specific people or stories. We reimagine this idea using both a mobile device and AR. Below, we outline key design considerations shaped by narrative theory [6].

C1. Embedding Time in Physical Space. Our system bridges the gap between abstract 2D representations and the way humans naturally interpret time by embedding timelines directly into the user’s physical environment as navigable spatial structures. This life-size visualization anchors time-based events in real-world

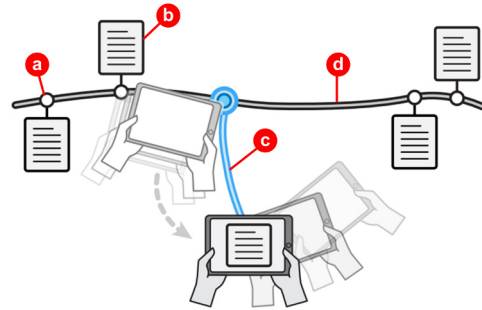


Figure 2: Key components of our timeline visualization:(a) Interactive nodes marking temporal positions, (b) Narrative cards displaying event/annotation details, (c) Connection lines anchoring cards to timeline positions, and (d) Structural lines defining the timeline’s form (curves, branches, etc.). This figure also illustrates card placement technique using the tablet’s bumping and dragging movement.

space, allowing them to reflect both chronological order and subjective understandings. Unlike screen-based tools that rely on panning and zooming with a mouse and keyboard, which can often disrupt temporal cognition [3], our spatial approach organizes events reflecting how people mentally structure and recall spatial sequences (e.g., remembering a day’s events in relation to physical locations).

C2. Rendering Flexible Timeline Structure. Our tool is designed to support the creation of flexible timeline structures, using a large AR canvas where users can start with rough event placements and refine them over time. Related items can be grouped together, important periods can expand to reflect their significance, and uneventful spans can contract to save space. This approach aligns with the concept of cartographic generalization used in map design [1]. This structure is equally well-suited for documenting past events, exploring complex histories, and imagining hypothetical futures.

C3. Differentiating Time Through Expressive Sculpting. While traditional tools limit expression to flat, 2D timelines, our system allows users to represent time in 3D space by directly shaping the timeline. Users can physically sculpt the timeline using simple spatial movements of the tablet [10]—drawing steep curves to represent turbulent or intense periods, and gentle slopes to indicate stable times. These expressive shapes remain visible throughout the authoring process, helping users convey meaning visually. Beyond shaping, users can also raise important moments vertically to show their narrative significance and use color-coded groups to differentiate between types of events or emotional tones (Figure 1).

C4. Event Layouts for Narrative Logic. The system allows users to organize time-based events in space according to their relationships or sequence, enabling the creation of a coherent story. For example, two timelines placed side by side can show events happening simultaneously or competing; a line that splits and re-connects may represent cause and effect; and nested lines can indicate that some events are part of larger ones. These structures can be changed as the user’s understanding evolves, allowing them to build timelines that are more structured and validated over time.

4 SYSTEM OVERVIEW

In this section, we detail the overall design of our authoring system.

4.1 Visualization Design

Our authoring tool is designed to support the creation of diverse and expressive timeline visualizations, composed of the following four visual components:

Structural lines form the deformable 3D backbone of a timeline, with their geometry reflecting narrative structures through curves, branches, and loops (Figure 2d). These pliable temporal

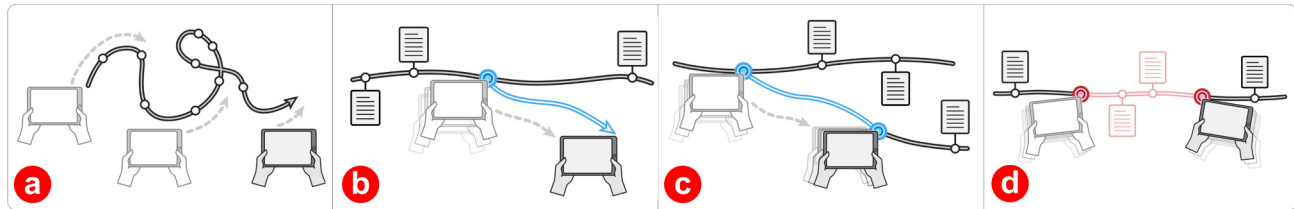


Figure 3: The AR timeline is shaped through a combination of physical tablet movements: (a) creating a line, (b) creating a branch, (c) connecting line segments, and (d) removing a line segment.

axes implement both spatiotemporal framing (C1) and flexible layouts (C4), allowing the timeline's shape to encode subjective interpretation of time through physical features such as curvature steepness and elevation (C3).

Interactive nodes function as temporal anchors along the structural lines, enabling precise placement and retrieval of cards via the tablet interface while preserving chronological order (C2) (Figure 2a). Vertical or spatial offsets indicate narrative importance through elevation and positioning.

Narrative cards link text and image data associated with events to spatial representation, displaying event details and annotations while maintaining a consistent orientation toward the user (Figure 2b). The card design supports visual differentiation (C3) through background color for event cards and text color for annotation cards, helping them quickly identify event or annotation types, emotional valence, and thematic groupings.

Connection lines provide visual feedback during card placement with dynamic, semi-transparent tethers between interactive nodes and cards (Figure 2c). These lines thicken and adopt the structural line's hue once placement is finalized, ensuring a clear visual association without interfering with ongoing interaction (C2).

4.2 Interaction Design

Our timeline authoring system transforms temporal representation through six core interaction techniques that allow users to shape the timeline using physical gestures. Leveraging tablet-AR hybrid interaction, the system supports direct manipulation of timeline geometry (linear paths, branches, connections), precise event/annotation management, and seamless content transfer between digital and physical spaces. The tablet serves not only as a spatial interface but also as a timeline data hub, allowing users to sort and select event data and access menus for choosing modes and features. Each feature follows embodied interaction principles [7], mapping device movement to temporal operations and turning abstract chronology into tangible, walkable structures.

Create a line. Users define the timeline's trajectories by physically moving the tablet through AR space, generating a persistent 3D line whose curvature reflects their movement patterns (C1). A two-finger tap gesture creates adjustable checkpoints along the curve (Figure 3a).

Create a branch. Once a primary timeline exists, users can create alternate narratives by bumping the tablet's corner against any point on the structural line in branch mode. This contact marks the divergence point, and subsequent tablet movements generate a new pathway extending from the original timeline (C4) (Figure 3b).

Connect line segments. For complex timelines, users can bridge separate timeline segments by bumping the tablet against a starting point to initiate a connection flow. As the tablet moves, a dynamic 3D tether stretches between the start and its current position, which users can sculpt into curves or angles in real time (C3, C4) (Figure 3c). The connection snaps into place when the tablet approaches the target segment, and the system automatically adjusts temporal properties to maintain chronological flow between the segments.

Remove line segment. Deletion employs a spatial selection method that preserves related content. In removal mode, users tap

the tablet's corner against both endpoints of the segment they want to delete (C1), creating a red-highlighted selection zone (Figure 3d). Event and annotation cards within the zone are also highlighted in red, previewing the content that will be transferred back to the tablet. A final two-finger tap confirms the deletion—removing the 3D line segment and transferring all associated events to a holding area on the tablet. This two-point selection helps prevent accidental deletions and keeps users aware of the original spatial context.

Bidirectional card transfer. On the tablet, users can create, sort, group, and edit event cards before transferring them to the timeline. To place content in AR, users first select an event or annotation card on the tablet in "To AR" mode. They then perform a bump gesture by tapping the tablet's corner against the desired timeline location, which snaps the event node to that point on the AR timeline. As the tablet moves away, a 3D connection line extends between the timeline and the event or annotation card displayed on the tablet, with its position determined by the tablet's final placement (C2). Cards remain oriented toward the user while staying anchored to their node connection (Figure 2).

For card retrieval in "To Tablet" mode, users retrieve cards by bumping the tablet against interactive nodes, triggering an animated transition that transfers the content back to the tablet interface. During transfers, the system preserves all temporal metadata, ensuring event cards can be precisely re-anchored elsewhere without losing their chronological relationships (C2). The physicality of these transfers mirrors real-world object handling, where the tablet becomes both storage medium and placement tool.

Reposition timeline. When timelines exceed the user's height or occupy substantial physical space (Figure 1), the system allows users to reposition oversized timelines along all three axes using directional controls (Move X+/-, Y+/-, Z+/-) with touch input. Using these controls, users can relocate the entire structure to a comfortable viewing position while preserving all temporal relationships within the timeline.

5 EXPLORATORY WORKSHOP

We conducted a workshop to explore the usability and expressive potential of our hybrid timeline authoring system. The goal is to understand how our system helps participants construct timelines in large space, including their use of free-form shaping, spatial event placement, and physical navigation, all supported by our system.

5.1 Participant

We recruited five male CS graduate students (P1–P5, aged 23–29), all familiar with creating timelines using free-form sketching tools such as MS PowerPoint, Goodnotes, or pen and paper. They reported varying levels of experience with AR/VR systems, including HoloLens and Meta Quest.

5.2 Apparatus and Prototype

The workshop was conducted in a 50'×30'×12' unobstructed lab space. For our hybrid timeline authoring system, we used a HoloLens 2 headset for spatial visualization and an 11-inch Samsung A9+ tablet, equipped with three flat infrared markers, for

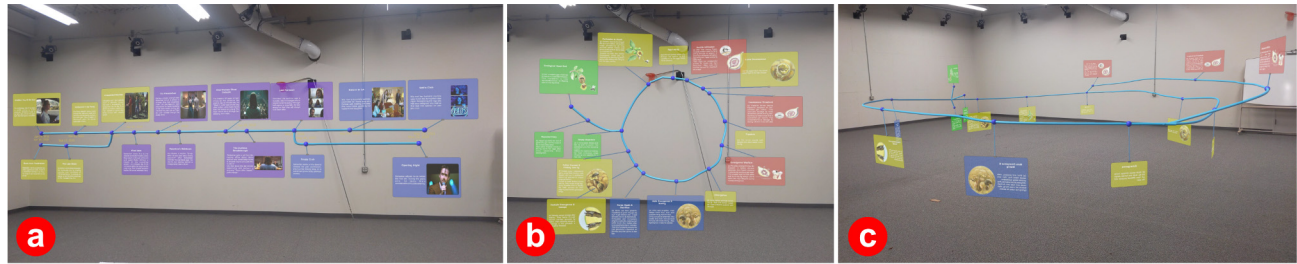


Figure 4: Participants' AR timeline for *La La Land* movie plot (a) and Fig wasp life cycle (b-c).

event selection and annotation management. The HoloLens's built-in AHAT sensor and Research Mode API were used to detect and track these IR markers. By retrieving the IR camera's intrinsic calibration parameters, we accurately projected the marker centroids into the HoloLens coordinate frame, establishing a unified spatial reference shared by both devices.

To enable low-latency, bidirectional synchronization of timeline data and user interactions, we implemented a peer-to-peer TCP socket architecture. The HoloLens app, built in Unity, uses an event-driven architecture where tablet interactions trigger updates to the scene and timeline data through modular components. The tablet runs a native web application developed with TypeScript and React Native, supporting cross-platform deployment.

5.3 Procedure and Data Collected

The workshop comprised three phases. In the pre-workshop phase, participants first completed informed consent procedures and demographic questionnaires, along with questions about their timeline design habits and prior AR/VR experience. This was followed by a 5-minute training session covering the core interaction techniques described in Section 4.2.

For the main authoring task, participants worked with two open-source datasets representing different temporal structures. The Fig wasp lifecycle dataset required representation of cyclical biological processes with potential branching outcomes, while the *La La Land* narrative dataset focused on linear storytelling with emotional arcs. During the 90-minute authoring sessions, participants actively engaged in timeline construction while moving through the space.

In the post-workshop phase, we collected data through structured questionnaires assessing tool usability, along with open-ended reflection prompts. Participants provided detailed feedback on their authoring experience, and the experiment proctor documented the final timeline designs through HoloLens screen captures.

5.4 Results and Observations

The workshop revealed that the dimensionality of the timeline fundamentally shapes users' physical engagement and memory strategies. Participants completed timeline authoring tasks in 40–90 minutes, with structural line editing proving challenging, while bidirectional card transfer was the most intuitive feature.

Our findings suggest that while a timeline's dimensionality (2D-like and 3D) does not affect how time is represented visually, it **plays a significant role in how users physically experience and remember the timeline during authoring and recall**. For the Fig wasp lifecycle dataset, implementations varied from upright 2D-like circles (P1, P3; Figure 4b) to 3D wraparound timelines that enabled rotational viewing (P4) (Figures 1a, 4c). Both approaches maintained the cyclical metaphor but differed in participants' spatial engagement and ways to leverage physical space—the 3D circular layouts acquired more physical space (Figure 4c), while the 2D-like versions (Figure 4b) enabled stationary visual overviews.

Emotional narratives leveraged vertical space as a mnemonic device. For the *La La Land* dataset, sloped timelines used height as emotional indicators: “peaks in in-love moments, valleys during

fight” (P2), “dramatic tension through tight curves . . . emotionally positive events for those placed high” (P1). Only P3 maintained a strictly horizontal linear progression (Figure 4a); however, they were aware that “the capability to [express emotional or dramatic aspects of timeline] was present with the tools I was given.”

Participant feedback suggested that **physical authoring aids temporal insight discovery**. P1 found “the order of off-branch events . . . was clearer after walking around,” while P3 gained awareness of parallel events by judging spacing. As P5 emphasized, 3D timelines “made us move and observe each point,” preventing the oversight risk of 2D displays. This spatial engagement actively reveals structural relationships like turning points and parallel events.

6 DISCUSSION AND FUTURE WORK

While our study demonstrates the potential of spatial timeline authoring, several limitations should be noted. The tablet-AR hybrid enables work at the user's personal comfortable positions like lap height, which may reduce fatigue compared to sustained mid-air gestures. However, prolonged use may still present ergonomic challenges. While preliminary feedback is positive, formal evaluation of optimal interaction heights and durations remains future work.

The lack of tactile feedback might have led to imprecise interactions during fine-grained editing, with one participant suggesting haptic cues for placement confirmation. Current card repositioning requires full removal (in “To Tablet” mode) and replacement (in “To AR” mode), which disrupts workflow continuity. We plan to address this by adding an in-place dragging feature to preserve spatial-temporal context during edits. In addition, our workshop was exploratory in nature and did not include a comparative study with screen-based authoring tools, leaving open questions about relative efficiency and learning curves.

Building on these insights, future research should explore the system's potential through different timeline analysis tasks to better characterize its strengths and limitations. The spatial and physical nature of the interface presents interesting opportunities for collaborative authoring scenarios that warrant further research. Additional studies involving domain experts from various fields could further validate the tool's practical applications while building on the creative authoring approaches observed in our current findings.

7 CONCLUSION

Our hybrid system transforms abstract temporal data into large-scale, expressive timelines, enabling emotionally meaningful event layouts in AR through users' physical movements and tablet interactions. Workshop findings showed that the system supported the creation of two main types of immersive timelines: 3D timelines that encouraged physical engagement, and 2D-like timelines that provided clearer overviews. This work offers foundational principles for immersive visualization authoring by combining spatial memory, embodied interaction, and personal interpretations of temporal events—moving beyond conventional screen-based authoring approaches.

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